CF03: Complete Formalism — A8 Scaling Theorem (Hierarchical Tachyonic Interfaces)

A Preprint

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Abstract

We formalize the VDM candidate axiom A8 ("hierarchical partition with logarithmic depth") for tachyonic interface-forming scalar-field systems by connecting three ingredients: (i) a phase-field (Ginzburg–Landau) energy with a tachyonic/double-well potential and a small interface width ε, (ii) its sharp-interface Γ-limit as a perimeter functional (Modica–Mortola), and (iii) a definition of *active hierarchy depth* $N(L)$ as the number of resolved interface scales between a macroscopic domain size L and a microscopic cutoff ℓ_0 (lattice spacing / minimum interface width). The Γ-limit yields boundary-energy concentration $E_{\text{exc}} \propto \text{Per}(A)$, implying the familiar interface-area scaling $E_{\text{exc}} = \Theta(L^{d-1})$ in d dimensions for bounded-shape partitions. We then show why $N(L)$ is at most logarithmic in L/ℓ_0 and provide a falsifiable interpretation of "hierarchy" in terms of a scale-by-scale *active interface census*. Finally, we define decisive gates that separate true log-depth organization from combinatorial blow-ups (e.g., 2^k interfaces at scale k) on lattice simulations.

Keywords: Γ-convergence; Modica–Mortola; phase-field; perimeter functional; mean curvature flow; microstructure scaling; hierarchy depth; tachyonic interfaces; Void Dynamics Model

1 Objective and scope

This CF document formalizes the VDM candidate axiom A8:[1]

> *In metriplectic scalar-field systems with tachyonic origin that admit pulled fronts with exponential tails, any finite-excess-energy large-domain trajectory must organize into a finite-depth hierarchical partition with logarithmic depth $N(L) = \Theta(\log(L/\ell_0))$, scale-gap separation, and nonzero boundary energy/information concentration fractions.*

The purpose here is **not** to assume hierarchy by analogy. Instead we provide a mathematically explicit bridge from (i) a tachyonic phase-field energy that forms interfaces, to (ii) a sharp-interface perimeter functional, and then to (iii) a precise notion of *hierarchy depth* that is provably bounded by a logarithm in the ratio L/ℓ_0.

Scope classification. This CF is **axiom-core** with respect to A8, but uses **derived-limit** results (Γ-convergence and mean-curvature limits of Allen–Cahn-type flows) as checks and as a source of falsifiable predictions.

What is being counted. Axiom A8 is about *depth* (how many distinct, resolved scales are simultaneously active), not about the raw number of pieces in a partition. The distinction matters: $N(L) = \Theta(\log L)$ is

[1] VDM canon files referenced throughout this paper are tracked as internal, versioned artifacts intended for Zenodo snapshots. See [Lietz, 2025a,b,c,d] for canonical registries and provenance conventions.

compatible with an *exponential* number of dyadic cells (2^k) at scale k, but it does *not* say the system must realize that combinatorial growth. The CF defines a measurable *active interface census* that makes the intended meaning falsifiable.

Minimal model. We analyze a standard phase-field energy

$$E_\varepsilon[\phi] \;=\; \int_\Omega \left[\frac{\varepsilon}{2}|\nabla\phi|^2 + \frac{1}{\varepsilon}W(\phi)\right] dx, \tag{1}$$

with a *double-well* potential W ("tachyonic at the origin" in the sense $W''(0) < 0$ with stable minima away from 0). This is the canonical interface-forming template behind Allen–Cahn dynamics and many microstructure models [Modica and Mortola, 1977, Modica, 1987, Dal Maso, 1993, Allen and Cahn, 1979, Desai and Kapral, 2009].

2 Claim inventory and decisive falsifiers

This section front-loads the claims and how they can fail.

2.1 Primary claims

Claim: C1: Sharp-interface bridge (Γ-limit) *Decisive metric:* Γ*-convergence residual in numerical recovery of perimeter scaling.*

The energy (1) Γ-converges, as $\varepsilon \to 0$, to a sharp-interface energy proportional to the perimeter of the phase boundary:

$$E_\varepsilon \xrightarrow{\Gamma} E_0 \equiv c_0\,\mathrm{Per}(A), \qquad A \equiv \{x \in \Omega : \phi(x) \approx +1\}, \tag{2}$$

with surface tension constant

$$c_0 = \int_{-1}^{1} \sqrt{2W(s)}\,ds. \tag{3}$$

This is the Modica–Mortola theorem and its refinements [Modica and Mortola, 1977, Modica, 1987, Dal Maso, 1993, Braides, 2002].

Claim: C2: Boundary-energy concentration *Decisive metric: fraction of total excess energy supported in an $O(\varepsilon)$ neighborhood of the interface.*

In the sharp-interface regime, the *excess* energy concentrates in a thin layer around the phase boundary; away from the boundary, $|\phi| \approx 1$ and energy density is asymptotically negligible. In measure terms, the energy densities converge to $c_0\,|D\chi_A|$, i.e., a surface measure supported on the interface [Modica, 1987, Dal Maso, 1993].

Claim: C3: Logarithmic bound on active hierarchy depth *Decisive metric: best-fit scaling exponent of $N(L)$ vs. $\log(L/\ell_0)$.*

Let L be the macroscopic diameter of Ω and ℓ_0 the smallest resolved length (lattice spacing / minimal interface width). Define $N(L)$ as the number of distinct length scales between L and ℓ_0 on which *active* interfaces are present (Definition 5). Then necessarily

$$N(L) \le \left\lceil \log_2\!\left(\frac{L}{\ell_0}\right) \right\rceil + 1. \tag{4}$$

Moreover, hierarchical constructions can realize $N(L) \ge c\log(L/\ell_0)$ with $c > 0$ in regimes where one interface per scale persists (§8).

Claim: C4: No combinatorial blow-up per scale (coarsening hypothesis) *Decisive metric: measured active-interface multiplicity per dyadic scale.*

For relaxation dynamics that reduce perimeter (Allen–Cahn/mean-curvature-type), same-scale boundaries tend to merge/annihilate, strongly suppressing exponential multiplicities $\sim 2^k$ at fixed scale k [Allen and Cahn, 1979, Bronsard and Kohn, 1991, Ilmanen, 1993, Huisken, 1984]. This is an *empirical* claim at the level of VDM lattices: it must be checked by a scale-by-scale boundary census.

2.2 Decisive gates

The following gates are designed so a notebook can certify or falsify the CF on real simulation outputs.

Gate: G1: Perimeter proxy scaling *Threshold: Fit* $E_{\text{exc}} \approx \sigma \widehat{\text{Per}}$ *with relative error* $\leq 5\%$ *over a scale sweep.*

Compute a discrete perimeter proxy $\widehat{\text{Per}}$ (§7). If E_{exc} does not scale linearly with $\widehat{\text{Per}}$ in the sharp-interface regime, the Modica–Mortola bridge is not the correct effective description.

Gate: G2: No per-scale exponential growth *Threshold: For each scale* k*, active multiplicity* m_k *satisfies* $m_k \leq m_{\max}$ *for a constant* $m_{\max}$ *independent of* k*.*

The CF permits many dyadic cells, but asserts that *active* boundaries do not appear with multiplicity $\propto 2^k$ at scale k under perimeter-reducing dynamics.

Gate: G3: Depth scaling and model selection *Threshold:* $N(L)$ *fits* $a\log(L/\ell_0) + b$ *with* $R^2 \geq 0.98$ *and beats power-law alternatives by AIC/BIC.*

Across a sweep in L at fixed microscopic cutoff ℓ_0, infer $N(L)$ from the active boundary census. Reject alternatives such as $N(L) \propto L^\alpha$ if information criteria prefer the log model.

Gate: G4: Dimensionless collapse *Threshold: Curves collapse when plotted vs.* L/ℓ_0 *(tolerance set by instrument resolution).*

Axiom A8 is a scaling claim: plots should collapse under the dimensionless ratio L/ℓ_0, not depend separately on L and ℓ_0.

3 Mathematical foundations

3.1 Phase-field energy and tachyonic potentials

Definition 1 (Double-well / tachyonic potential)**.** A function $W : \mathbb{R} \to [0, \infty)$ is called a *double-well* potential with wells at ± 1 if: (i) $W(\pm 1) = 0$, (ii) $W(s) > 0$ for $s \neq \pm 1$, (iii) W is smooth in a neighborhood of $[-1, 1]$, and (iv) $W''(0) < 0$ (unstable origin). A standard example is $W(s) = \frac{1}{4}(1 - s^2)^2$, for which $c_0 = 2\sqrt{2}/3$ in (3).

The small parameter ε in (1) sets the interface thickness. In lattice simulations, the smallest meaningful scale is $\ell_0 \sim \max\{\varepsilon, \Delta x\}$, where Δx is the mesh spacing.

Connection to VDM axioms. The VDM master evolution law is metriplectic, schematically $\dot{q} = J\,\delta\mathcal{I}/\delta q + M\,\delta\Sigma/\delta q$ with degeneracy constraints (**VDM-A-4**–**VDM-A-5**). In scalar-field specializations, $\mathcal{I}$ typically contains a gradient penalty and a tachyonic potential, of which (1) is the standard sharp-interface scaling. This CF does not assume a specific (J, M) beyond the existence of perimeter-reducing relaxation in the dissipative sector.

3.2 Γ-convergence and the Modica–Mortola theorem

We recall the definition only at the level needed for the result. For full details see [Dal Maso, 1993, Braides, 2002].

Definition 2 (Γ-convergence (informal))**.** Let F_ε be a family of functionals on a topological space X. We say F_ε Γ-converges to F_0 if: (i) for every $u_\varepsilon \to u$, $\liminf_{\varepsilon\to 0} F_\varepsilon(u_\varepsilon) \geq F_0(u)$ (liminf inequality), and (ii) for every u there exists $u_\varepsilon \to u$ such that $\limsup_{\varepsilon\to 0} F_\varepsilon(u_\varepsilon) \leq F_0(u)$ (recovery sequence).

Theorem 1 (Modica–Mortola)**.** *Let* $\Omega \subset \mathbb{R}^d$ *be bounded and let* W *be a double-well potential. Then the phase-field energies* (1) *Γ-converge in* $L^1(\Omega)$ *to*

$$E_0[\chi_A] = c_0\,\text{Per}(A), \tag{5}$$

where $A \subset \Omega$ *is a set of finite perimeter (Caccioppoli set) and* c_0 *is given by* (3).

Remark 1 (Interpretation)*.* Theorem 1 states that, in the sharp-interface regime, the only surviving energetic cost is the area/perimeter of the phase boundary. This is the precise sense in which interface energy is *surface-dominated*.

4 Boundary-energy scaling and interface measures

4.1 Interface-area scaling $E_{\mathrm{exc}} = \Theta(L^{d-1})$

Assume Ω has characteristic diameter L (e.g., a cube $[0, L]^d$) and that the limiting set A has a *bounded-shape* interface: the total interface area scales like L^{d-1}, as it does for partitions with $O(1)$ macroscopic components. Then Theorem 1 gives

$$E_{\mathrm{exc}} \approx c_0 \,\mathrm{Per}(A) = \Theta(L^{d-1}). \tag{6}$$

This scaling is a key component of the VDM A8 narrative: "excess energy concentrates at boundaries."

4.2 Compatibility with logarithmic hierarchy depth

Logarithmic depth does *not* imply super-surface energy. A representative hierarchy model is: at dyadic level k, boundaries have characteristic area proportional to $L^{d-1}2^{-k(d-1)}$. Then the total perimeter is a geometric sum:

$$\mathrm{Per}(A) \;\sim\; \sum_{k=0}^{N(L)-1} L^{d-1}2^{-k(d-1)} \;\le\; \frac{1}{1-2^{-(d-1)}}\,L^{d-1} \qquad (d \ge 2), \tag{7}$$

so E_{exc} remains $\Theta(L^{d-1})$ even if $N(L) = \Theta(\log L)$. In $d = 1$, Per counts boundary points; see §8.

5 Hierarchy depth and its logarithmic bound

5.1 Definition of active depth

The CF needs a definition that is both conceptually faithful to A8 and implementable on a lattice.

Definition 3 (Dyadic scale ladder)**.** Fix L and a microscopic cutoff $\ell_0 > 0$. Define dyadic scales

$$\ell_k \equiv 2^{-k}L, \qquad k = 0, 1, 2, \ldots \tag{8}$$

and define $K_{\max}(L)$ as the largest integer with $\ell_{K_{\max}} \ge \ell_0$. Equivalently, $K_{\max}(L) = \lfloor \log_2(L/\ell_0)\rfloor$.

Definition 4 (Active interface at scale ℓ_k)**.** Let ϕ be a phase field on a discrete lattice. Let $\mathcal{B}(\phi;\ell_k)$ be a procedure that extracts a set of "interfaces visible at scale ℓ_k" (§7). We say ℓ_k is *active* if $\mathcal{B}(\phi;\ell_k)$ is nonempty after subtracting interfaces already visible at coarser scales (a "new boundary" criterion).

Definition 5 (Active hierarchy depth)**.** The *active hierarchy depth* $N(L)$ is the number of active scales in the dyadic ladder between L and ℓ_0:

$$N(L) \equiv \#\{k \in \{0, \ldots, K_{\max}(L)\} :\ \ell_k \text{ is active}\}. \tag{9}$$

5.2 Upper bound proof

Proposition 1 (Logarithmic upper bound)**.** *For any definition of activity based on the dyadic ladder, $N(L) \le K_{\max}(L) + 1$, hence* (4)*.*

Proof. There are only $K_{\max}(L) + 1$ dyadic scales $\ell_k \ge \ell_0$. Since $N(L)$ counts a subset of those scales, the bound is immediate. ☐

Remark 2 (Why this is not vacuous)*.* The bound becomes meaningful only once "active" is operationalized (Definition 4) and measured. The gate **G3** does exactly this: it tests whether empirical $N(L)$ actually grows like $\log(L/\ell_0)$, rather than saturating or growing as a power law.

6 Perimeter reduction and suppression of combinatorial blow-up

A8 is stronger than the trivial counting bound: it asserts the system *organizes into a hierarchy* rather than producing a combinatorial number of same-scale boundaries. This section clarifies the mechanism that can enforce the suppression.

6.1 Allen–Cahn / mean-curvature limit as a perimeter-reducing model

The *gradient flow* of (1) in L^2 is the Allen–Cahn equation

$$\partial_t \phi = \varepsilon \Delta \phi - \frac{1}{\varepsilon} W'(\phi), \tag{10}$$

introduced to model curvature-driven interface motion [Allen and Cahn, 1979]. A key fact is that, in the sharp-interface limit, the phase boundary evolves by mean curvature flow (in suitable weak senses) [Bronsard and Kohn, 1991, Ilmanen, 1993]. Mean curvature flow decreases surface area; in particular, for smooth convex hypersurfaces it drives the surface to a sphere while strictly decreasing area [Huisken, 1984].

Implication for hierarchy. If the dissipative sector of a VDM lattice implements a perimeter-reducing relaxation analogous to (10), then same-scale interfaces are unstable: they merge and annihilate on a timescale proportional to their separation. Therefore, at any given resolved scale, one expects only *O(1) persistent active interfaces* rather than $\Theta(2^k)$ interfaces per dyadic level. This is precisely Claim C4 and Gate G2.

6.2 Relation to microstructure scaling literature

A broad calculus-of-variations literature studies how competing bulk and surface terms select hierarchical microstructures and scaling laws; canonical examples include martensitic microstructures with branching patterns [Kohn and Müller, 1994, Conti, 2000]. The present CF does *not* import those models as axioms; rather, it uses them as corroborating evidence that (i) hierarchical patterns are natural minimizers under surface/bulk competition and (ii) scaling laws are testable by energy and interface observables.

7 Discrete observables and CFN extraction protocol

This section specifies what a notebook must compute to evaluate gates **G1**–**G4**.

7.1 Perimeter proxy

On a lattice, two common proxies for interface measure are:

1. **Gradient measure:** $\widehat{\mathrm{Per}} \equiv \sum_x |\nabla\phi(x)|\,\Delta x^d$. In the sharp-interface regime $|\nabla\phi|$ concentrates near boundaries.
2. **Sign-change measure (binary proxy):** If ϕ is approximately ± 1, threshold $\chi(x) = \mathbf{1}_{\phi(x)>0}$ and count boundary edges in the grid graph.

Either is acceptable provided it passes a calibration test on known shapes (circle/sphere, stripe), and provided **G1** is evaluated using the same proxy.

7.2 Scale-by-scale active boundary census

Define a multiscale coarse-graining operator $\mathcal{C}_{\ell_k}$ (e.g., Gaussian smoothing with width ℓ_k followed by downsampling). At each scale:

1. compute the coarse-grained field $\phi_k \equiv \mathcal{C}_{\ell_k}[\phi]$;
2. extract a boundary set $\mathcal{B}_k \equiv \mathcal{B}(\phi_k; \ell_k)$ (e.g., $|\phi_k| < \tau$ or boundary edges of thresholded χ_k);
3. define *new* boundaries $\Delta\mathcal{B}_k \equiv \mathcal{B}_k \setminus \mathrm{dilate}(\mathcal{B}_{k-1})$ to avoid double-counting across scales;
4. measure multiplicity m_k (connected components or total length/area of $\Delta\mathcal{B}_k$).

Then $N(L) = \#\{k : m_k > 0\}$ and **G2** checks whether m_k remains bounded in k.

7.3 Model selection for $N(L)$

To implement **G3**, perform a sweep over L at fixed ℓ_0 and compute $N(L)$ for each run. Fit $N(L) = a\log(L/\ell_0) + b$ and compare to power-law alternatives $N(L) = c(L/\ell_0)^\alpha$ using information criteria.

8 Worked example: 1D hierarchical interfaces

The purpose of this example is to make the "depth" concept concrete.

8.1 Single interface energy

For $W(s) = \frac{1}{4}(1-s^2)^2$, the surface tension constant is

$$c_0 = \int_{-1}^{1} \sqrt{\tfrac{1}{2}(1-s^2)^2}\, ds = \frac{2\sqrt{2}}{3}. \tag{11}$$

In one dimension, the sharp-interface energy is $E_0 = c_0 \operatorname{Per}(A)$, where $\operatorname{Per}(A)$ is the number of boundary points of $A \subset \mathbb{R}$. Thus each additional interface adds $\approx c_0$ energy in the sharp-interface regime.

8.2 Nested construction and log scaling

Place one interface at each dyadic scale $\ell_k = 2^{-k}L$ until $\ell_k \approx \ell_0$. Then

$$N(L) \approx \left\lfloor \log_2\left(\frac{L}{\ell_0}\right)\right\rfloor + 1, \qquad E_0 \approx c_0\, N(L) = \Theta\left(\log\left(\frac{L}{\ell_0}\right)\right). \tag{12}$$

This is not a claim about typical dynamics; it is a realizability witness showing that log-depth hierarchies are consistent with finite excess energy.

9 Limitations and future work

- **Rigorous vs. empirical.** The Γ-limit (Claim C1) is rigorous under standard hypotheses [Modica and Mortola, 1977, Modica, 1987, Dal Maso, 1993]. The suppression of per-scale exponential multiplicity (Claim C4) depends on the actual dissipative dynamics and must be validated on VDM lattice runs.
- **Definition of "active".** Definition 4 is intentionally operational; different extraction operators $\mathcal{B}$ can change $N(L)$ at finite resolution. **G4** is intended to detect such artifacts via dimensionless collapse.
- **Beyond scalar phase fields.** A8 refers to a class of metriplectic scalar-field systems with pulled fronts and exponential tails. Extending the CF to coupled fields or nonlocal terms requires repeating the sharp-interface and coarsening checks.

10 Provenance and reproducibility

Commit: `91332f4cf89a52d239eaab6659b8df8d8b9f3202` **Seed(s):** `N/A` **Artifacts:** `Source: Derivation/Complete-Formalisms/CF03_A8_Scaling_Hierarchical_Interfaces.md (SHA256: 0ff065faf617040d7a44df4869acede28a026f2f60b8ddb678b674fbf52b52db)`

Reference implementations:

- Notebook: `Derivation/Notebooks/01_Scale/CF3_A8_Scaling_Hierarchical_Interfaces.ipynb`
- A8 preprint context: `T8_A8_Lietz_Infinity_Resolution_Conjecture.pdf` [Lietz, 2025e]

A Minimal pseudocode: active boundary census

```
# Inputs: field phi on grid, domain size L, cutoff ell0
# Output: depth N and per-scale multiplicities m_k

scales = []
k = 0
while (L / (2**k)) >= ell0:
    scales.append(L / (2**k))
    k += 1
```

```
B_prev = empty_set()
N = 0
multiplicities = []

for ell in scales:
    phi_c = coarse_grain(phi, width=ell)
    B = extract_boundaries(phi_c)          # e.g. edges of thresholded sign
    dB = B - dilate(B_prev)                # new boundaries at this scale
    m  = count_components(dB)              # or measure length/area

    multiplicities.append(m)
    if m > 0:
        N += 1

    B_prev = B

return N, multiplicities
```

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